

Sad or hopeless & teen smoking

Group SadOrHopeLess × Response CurrentCigaretteUse

THE METHOD

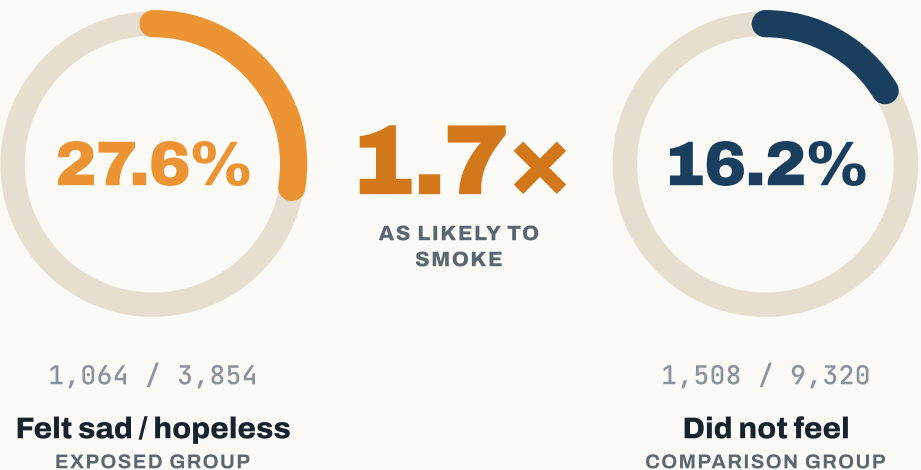
Two-proportion z-test

$H_0: p_{\text{yes}} = p_{\text{no}}$ · $H_1: p_{\text{yes}} \neq p_{\text{no}}$
 $\alpha = 0.05$ · two-sided · $\text{diff} = p_{\text{yes}} - p_{\text{no}}$

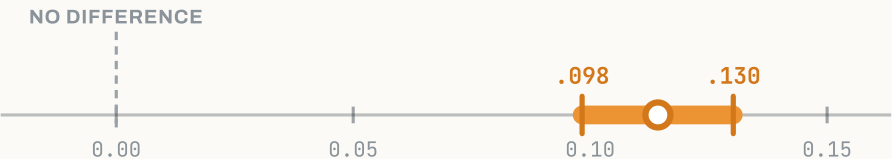
MAIN RESEARCH QUESTION · Two-proportion z-test

Do students who feel sad or hopeless smoke more?

01 Current cigarette use by group



02 Difference in proportions 95% confidence interval ($p_{\text{yes}} - p_{\text{no}}$)



diff +0.114

z 15.05

p < .0001

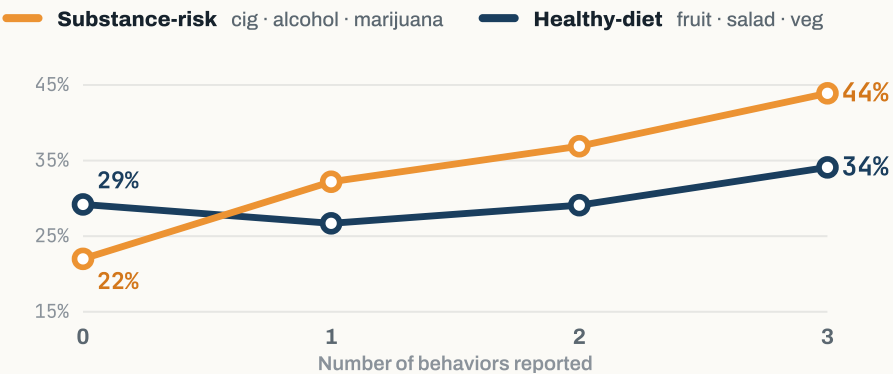
$\alpha = .05$ Reject H_0

Current smoking is **significantly higher** with distress — a +11.4 pp gap, with the 95% CI entirely above zero.

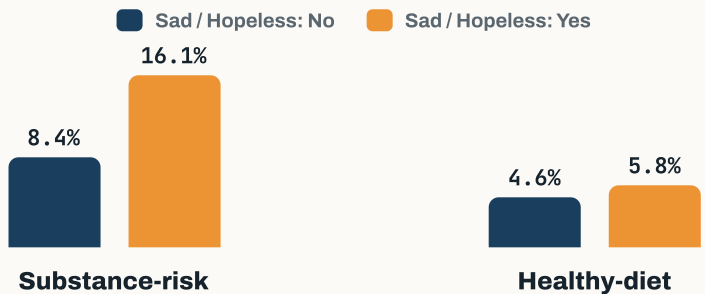
ADDITIONAL EDA QUESTION · Exploratory

Do risky behaviors cluster with distress more than healthy ones?

01 Sad / hopeless % as behavior count rises (0—3)



02 Reporting all three behaviors at once



Risk behaviors **nearly double** with distress — healthy habits stay flat.